

PROJECT REPORT

Metro Ticket Generating System in ServiceNow

1. INTRODUCTION

1.1 Project Overview

The Metro Ticket Generating System is a ServiceNow-based digital ticket-booking application. It uses the ServiceNow Service Portal and a Service Catalog item called “Book A Metro Ticket”. A passenger can select Starting From and Going To stations, choose Single Journey or Return Journey, select the number of passengers, view the automatically calculated fare, select a payment mode and generate a digital QR ticket.

1.2 Purpose

The purpose is to make metro ticket booking simple, convenient and automated, reducing manual effort, waiting time and fare-calculation errors while providing a digital QR ticket.

2. IDEATION PHASE

2.1 Problem Statement

Metro passengers may face queues, manual effort and confusion while selecting routes and calculating fares. A simple digital system is needed to bring ticket-booking activities into one convenient flow.

2.2 Empathy Map Canvas

Area	Passenger Insight
Says	I want to book my ticket quickly and know the correct fare.
Thinks	How long will booking take? Is the fare correct?
Does	Selects stations, journey type, passengers, payment mode and submits.
Feels	Before: rushed/confused. After: relieved, confident and satisfied.
Pains	Queues, manual fare calculation and multiple booking steps.
Gains	Quick booking, automatic fare calculation and digital QR ticket.

2.3 Brainstorming

- ServiceNow Service Portal ticket booking
- Metro Station Detail reference table
- Single/Return Journey options
- Automatic route-based fare calculation
- UPI, CARD and Others payment modes
- Digital ticket and QR-code generation
- Mandatory-field and input validation

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Passenger Action	Expected Result
1	Open ServiceNow Service Portal	Access booking service
2	Select Book A Metro Ticket	Booking form opens
3	Select source and destination	Route selected
4	Select journey type and passengers	Journey details recorded
5	View automatic fare	Correct fare displayed
6	Select payment mode	Payment choice recorded
7	Submit request	Booking is created
8	Generate QR ticket	Digital ticket is displayed

3.2 Solution Requirement

ID	Requirement	Description
FR-1	Station Selection	Starting From and Going To station selection
FR-2	Journey Selection	Single Journey / Return Journey
FR-3	Passenger Selection	Number of passengers
FR-4	Fare Calculation	Automatic fare based on route, journey and passengers
FR-5	Payment Mode	UPI / CARD / Others
FR-6	Ticket Submission	Submit and store booking request
FR-7	Digital Ticket	Display ticket details
FR-8	QR Ticket	Generate and display QR code
FR-9	Validation	Validate required/valid inputs

Non-functional requirements: Usability, Reliability, Performance, Security and Scalability.

3.3 Data Flow Diagram

Metro Passenger → ServiceNow Service Portal → Book A Metro Ticket → Validation & Fare Calculation → Metro Station Detail / ServiceNow Ticket Data → QR Generation → Digital Metro QR Ticket.

3.4 Technology Stack

Area	Technology
User Interface	ServiceNow Service Portal
Application	ServiceNow Service Catalog
Logic	Catalog Client Scripts / JavaScript
Data	ServiceNow tables; Metro Station Detail (u_metro_station_detail)
QR	QR Code Generation API used by the project
Hosting	ServiceNow Cloud Platform
ML	Not applicable

4. PROJECT DESIGN

4.1 Problem Solution Fit

- Customer Segment: metro passengers, students, employees and daily/occasional commuters.
- Constraints: queues, limited time, manual fare calculation and multiple steps.
- Jobs-to-be-Done: book ticket, select route, calculate fare, choose payment and receive ticket.

- Root Cause: manual or multi-step ticketing process.
- Triggers: need to travel, rush hours and desire to avoid queues.
- Emotions: before – rushed/confused; after – relieved and satisfied.
- Solution: ServiceNow digital booking with automatic fare calculation and QR ticket.

4.2 Proposed Solution

A ServiceNow-based digital metro ticket system where passengers enter journey details, receive automatic fare calculation, choose a payment mode, submit the booking and receive a digital QR ticket.

4.3 Solution Architecture

Passenger → ServiceNow Service Portal → Book A Metro Ticket Catalog Item → Catalog Client Scripts → Metro Station Detail / Ticket Records → QR Generation → Digital QR Ticket.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Epic	User Stories	Points
Sprint-1	Station Setup	USN-1, USN-2	3
Sprint-1	Journey Details	USN-3	3
Sprint-1	Passenger Details	USN-4	3
Sprint-1	Fare Calculation	USN-5	3
Sprint-2	Payment	USN-6	2
Sprint-2	Digital Ticket	USN-7	3
Sprint-2	QR Ticket	USN-8	3
Sprint-2	Validation	USN-9	2
Sprint-2	ServiceNow Portal	USN-10	5
Sprint-2	Testing & Refinement	USN-11	5

Sprint 1 = 12 Story Points | Sprint 2 = 20 Story Points | Total = 32 Story Points | Planned Velocity = $32 / 2 = 16$ Story Points per Sprint.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project is rule-based and does not contain a machine-learning model. Application-level functional and performance testing is therefore used.

- Station selection validation
- Journey type validation
- Passenger count validation
- Automatic fare calculation
- Payment-mode selection
- Ticket submission
- QR-ticket generation
- Response-time checking
- QR API response checking

7. RESULTS

7.1 Output Screenshots

Insert actual project screenshots showing the following outputs:

- Book A Metro Ticket form
- Starting From / Going To selection
- Journey type and passenger count
- Automatic fare calculation
- Payment mode
- Digital QR ticket

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Simple digital booking
- Reduces manual effort and waiting
- Automatic fare calculation
- Multiple payment-mode options
- Digital QR ticket
- Can be extended with more stations/routes

Disadvantages:

- Depends on the ServiceNow platform
- Current implementation covers configured stations/routes
- QR generation depends on the configured QR service
- Real payment-gateway processing is not defined in the current project
- Live train status and advanced analytics are not included

9. CONCLUSION

The Metro Ticket Generating System demonstrates how ServiceNow Service Portal, Service Catalog, custom tables and Catalog Client Scripts can be combined to create a convenient digital metro-ticket booking flow with automatic fare calculation and QR-ticket generation.

10. FUTURE SCOPE

- Real payment-gateway integration
- More stations and dynamic fare management
- Ticket history, cancellation and refunds
- Live train status and alerts
- Email/SMS notifications
- QR validation at entry points
- Administrator dashboards and analytics
- Enhanced role-based security

11. APPENDIX

Source Code

The project logic is implemented in ServiceNow using Catalog Client Scripts and platform configuration. The main logic covers validation, station/journey handling, passenger count, fare calculation and QR-ticket flow.

Dataset Link

No external public dataset is used. Metro station information is maintained in the custom ServiceNow table Metro Station Detail (u_metro_station_detail).

GitHub & Project Demo Link

GitHub Link: <https://github.com/yadalapujitha05/Metro-Ticket-Generation>

Project Demo Link: <https://drive.google.com/file/d/1l2WWvJeYDzge2aeepf56gZIOM3KciWkU/view?usp=sharing>

Add the actual repository and ServiceNow demo links here if available.